# AUTO: before the match

Team 37102 · BioBuzz AUTO RED / BLUE (Pedro) · 30 s, hands off

**1 🤝 Talk to partner**
Who has S1, who parks in A / B, the GARDEN, our lane (yellow box).

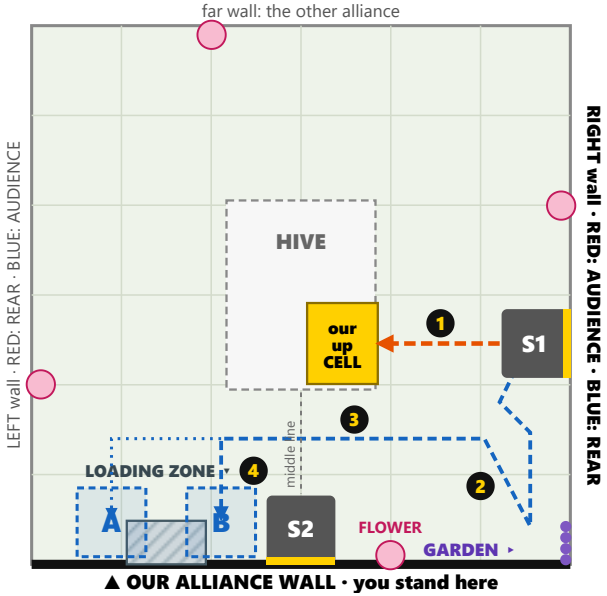
**2 📋 Pick op mode**
**BioBuzz AUTO RED** / **BLUE** = our colour (no other AUTO). TeleOp is preselected.

**3 📍 Place + load**
Robot at **S1** or **S2** (map). **4 pre-loads**, one in each windmill pocket.

**4 ✋ INIT, hands off**
Don't touch the robot until START: the Pinpoint calibrates while it is still.

**5 🎮 Check + START**
Set gamepad 1 buttons, read the screen. ▶ **START** when the match starts.

## 📍 Where to place it




▲ **OUR ALLIANCE WALL · you stand here**

**Seen from where you stand: the same for RED and BLUE.** Yellow bar = intake (front); launcher + camera at the back. **1** shoot · **2** GARDEN · **3** lane beside the HIVE · **4** park in B (dotted: to A)

**S1 · CELL wall** (default, best): **intake against the wall on your RIGHT** (RED: audience wall, BLUE: rear wall), **centred on our up CELL** (robot middle in line with the CELL middle), square to the wall. It shoots from here.

**S2 · ALLIANCE wall: intake against our wall, centred on the middle line** (between the FLOWER and the LOADING ZONE), square to the wall. It drives about 5 s to the shot spot, so the GARDEN is short or skipped.

## 🎮 INIT buttons (gamepad 1)

| Press | Changes | Default |
|---|---|---|
| **D-pad ← or →** | Start S1 ⇄ S2 | **S1** |
| □ **Square** | Park slot B ⇄ A | **B** |
| ✕ **Cross** | Shoot pre-loads | **ON** |
| △ **Triangle** | GARDEN | **ON** |
| ○ **Circle** | Stretch | **OFF** |
| **D-pad ↑ / ↓** | Start delay +1 / −1 s | **0 s** (0–10) |
| **R1 / L1** | Park by +1 / −1 s | **24 s** (18–27) |

✕ △ ○ switch ON ⇄ OFF. One press = one change. **Start delay**: it waits that long after START. **Park by**: the latest time it stops and goes to park. **Stretch**: shoot again after the GARDEN (page 2). The screen calls L1 / R1 "LB / RB". **Every INIT starts from these defaults**: after a new INIT, set them again.

### 🤝 Agree with your partner (every match)

1. **Start:** only one robot can use S1.
2. **Park:** one robot in A, one in B (the screen says `we park in slot B, they take slot A`). They park **only at their end** of the zone, close to the wall.
3. **GARDEN:** we take our GARDEN POLLEN. They should not.
4. **Lane:** at the end of AUTO they stay out of our lane beside the HIVE (map, 3).

B is the default: the way to A goes past B.

## 📺 INIT screen: a good one

```
BIOBUZZ AUTO RED - Pedro                    our colour?
Start (D-pad L/R) : CELL wall: side wall in front of our up   as agreed
CELL (shoots from here)
Park slot (Square/X) : B = MIDDLE end of the LOADING          as agreed
ZONE (toward the FLOWER)
>> Tell your partner: we park in slot B, they take slot A
Shoot pre-loads (Cross/A) : ON                          ON
GARDEN POLLEN (Triangle/Y) : ON                         ON
Stretch: shoot again (Circle/B) : OFF
Start delay (D-pad U/D) : 0 s
Park by (LB/RB) : 24 s at the latest
Plan : shot spot + 4 pre-loads done ~6 s
Plan : GARDEN done ~17 s (intake on, 2 in back and forth ...)
Plan : PARK B starts by 23 s, takes ~6 s
Route check : OK                                        must be OK
Pedro : ready                                           must be ready
Camera : no CELL in view (normal at the wall)           ok
Place: intake against the AUDIENCE wall, centred on           robot there?
our up CELL, square to the wall
```

RED, all defaults. BLUE: AUTO BLUE, REAR wall. Plan times are estimates.

| Bad line | What happens · what you do |
|---|---|
| `Pedro : NOT READY …` | No Pinpoint or Pedro not tuned: AUTO only backs 8 in off the wall (LEAVE, 3 points). Keep still; tell the coach. |
| `Camera : NOT FOUND …` | Fires the pre-loads unaimed, no stretch. Tell the coach. |
| `GARDEN POLLEN : ON but needs shooting (G407)` | Shoot is OFF, so no GARDEN. ✕ turns Shoot ON. |
| `Route check : PROBLEM …` | Tell the coach. |
| `GARDEN ON but no "GARDEN done" Plan line` | Not enough time: it will skip the GARDEN. |

# AUTO: in the match

what the robot does · hands off · if something goes wrong · checklist

## What the robot does by itself (30 s)

**1 0–1 s**
**LEAVE**
Leaves the wall, spins up the flywheel.
**LEAVE +3 (counted at the end)**

**2 1–7 s**
**SHOOT**
Camera aims at our up CELL, fires the 4 pre-loads.
**HIVE TIP +20 if it tips**

**3 7–14 s**
**GARDEN**
Sweeps slowly along the wall into the GARDEN corner, intake on.
**POLLEN for TeleOp**

**4 14–18 s**
**CORNER**
Intake on up to 4 s, going 2 in back and forth, so the last POLLEN rolls in.

**5 ~18 s**
**PARK**
In our LOADING ZONE. Slot B: parked by ~24 s. Slot A: by ~28 s.
**PARK +5**

**STRETCH (○)**
**OFF by default**
Back to the spot, fires the GARDEN POLLEN only if the camera sees our CELL still **up**.

Times are from S1 with no start delay. A start delay makes it all later. From S2 it first drives about 5 s to the shot spot.

**PARKING ALWAYS WINS: at the Park-by time (24 s, or earlier if the drive is long) it stops what it is doing and parks.**

## During AUTO: hands off

- **Don't touch** the robot or the gamepads. The gamepads do nothing in AUTO.
- **Don't press STOP.** All motors stop at once and the robot won't park. Emergencies only.
- It waits or skips a step? **Let it**: those are its own rules (middle column). After the match, tell the coach: the log has every step.
- Watch the `Step` line to see what it is doing.

## After AUTO: TeleOp

1. The 30 s AUTO timer ends AUTO. The robot holds its park until then.
2. **BioBuzz TeleOp** is already selected for you: press **INIT**.
3. `Alliance` line: AUTO already set our colour. Check it anyway.
4. Field-centric on? The screen says `Start heading: from AUTO`. Don't press OPTIONS.
5. Don't touch the robot. ▶ **START** when TeleOp starts.

## If this happens in AUTO

| If... | the robot... |
|---|---|
| **No CELL in view** | fires the pre-loads on the planned heading. |
| **Our CELL is already down (HIVE tipped)** | never fires into it (G417). No GARDEN; it parks. |
| **Windmill not empty after shooting (e.g. a jam)** | skips the GARDEN (G407: max 4 balls) and parks. |
| **Blocked 1 s before the park (e.g. by our partner)** | drops that step and goes to park. |
| **Blocked while parking** | waits and tries again until AUTO ends (stays if partly in the zone). |

**Practice:** set our HIVE to the match start first: **our CELL up** (the CELL that points at a FLOWER is the down one). If our CELL is down, **AUTO won't shoot** (G417).

## The Step line during AUTO

| Step says | It is... |
|---|---|
| `DELAY` | waiting (start delay) |
| `SHOT SPOT · AIM … · FIRE …` | shooting the pre-loads |
| `GARDEN …` | in the GARDEN |
| `PARK: …` | driving to the slot |
| `PARKED B / PARKED A` | **done ✓** |
| `FALLBACK DONE (LEAVE only)` | Pedro was not ready |

## PRE-MATCH CHECKLIST

- ☐ **Partner agreed:** who has S1 · A or B · our GARDEN · our lane
- ☐ **Op mode:** BioBuzz AUTO RED / BLUE = our colour
- ☐ **4 pre-loads**, one in each pocket
- ☐ **Robot at S1 or S2:** intake against the wall, centred, square
- ☐ **INIT**, then hands off the robot
- ☐ Title line = **our colour**
- ☐ **Start** and **Park slot** as agreed
- ☐ **Shoot ON, GARDEN ON** · stretch, delay, park by as agreed
- ☐ **Route check: OK** · **Pedro: ready** · Camera not NOT FOUND
- ☐ **Place** line = where the robot is
- ☐ Last look, then ▶ **START** when the match starts. Hands off.